



GIET UNIVERSITY, GUNUPUR
SCHOOL OF ENGINEERING AND TECHNOLOGY
Department of CSE (AIML)
Database Management Systems (DBMS) — 3rd Semester
ASSIGNMENT — II

Submission Date:- 17th September 2026

Unit II Coverage:

Relational Model and its terminologies, Relational Database Constraints; Mapping ER model to relational schema; Relational Algebra (Selection, Projection, Union, Intersection, Set Difference, Cartesian Product, Rename, Join, Division); SQL — DDL, DML, DQL (SELECT, WHERE, ORDER BY, GROUP BY, Aggregate Functions, Joins); Tuple Relational Calculus; Domain Relational Calculus.

SECTION A (Short Answers)

1. Define a relation, tuple, and attribute in the relational model.
2. What is the difference between a candidate key and a primary key?
3. Define referential integrity constraint with an example.
4. What is a foreign key? Why is it needed?
5. Differentiate between entity integrity and domain constraint.
6. Define degree and cardinality of a relation.
7. What is the Selection (σ) operator in relational algebra? Give its syntax.
8. What is the Projection (π) operator? How is it different from Selection?
9. State the condition required for the Union and Intersection operations to be valid (Union Compatibility).
10. Differentiate between Set Difference and Intersection in relational algebra.
11. Differentiate between DDL, DML, and DQL with one command example each.
12. What is the difference between DELETE, DROP, and TRUNCATE?
13. Define Tuple Relational Calculus (TRC) and give its general syntax form.
14. Define Domain Relational Calculus (DRC) and how it differs from TRC.
15. Differentiate between Left Outer Join and Right Outer Join with a one-line example.

SECTION B

1. Explain Union, Intersection, and Set Difference operations in relational algebra with union-compatibility conditions and examples.
 2. Write SQL DDL commands to CREATE a table "Employee" (EmpID, Name, Dept, Salary) with Primary Key, Not Null, and Check constraints. Also show ALTER (add a column) and DROP TABLE syntax.
 3. Write SQL DML commands to INSERT five records into a table, UPDATE a column based on a condition, and DELETE records satisfying a given condition.
 4. Write short Domain Relational Calculus (DRC) expressions for two sample queries and briefly compare DRC with TRC.
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SECTION C

1. Discuss in detail the step-by-step procedure for mapping an ER model to a Relational Schema. Explain how strong entities, weak entities, and relationship sets (1:1, 1:N, M:N) are mapped, with examples.
2. Explain SQL Join operations in detail with syntax and examples: INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, and FULL OUTER JOIN, using two related tables (e.g., "Student" and "Department").
3. Consider the schema: Student(RollNo, Name, DeptID), Department(DeptID, DeptName), Enrollment(RollNo, CourseID, Marks). *[10 Marks]*
 - a) Write relational algebra expressions for: (i) students enrolled in a given course, (ii) students who scored above 80 in all courses (division), (iii) natural join of Student and Department.
 - b) Write equivalent SQL queries for (i) and (ii), and also write TRC/DRC expressions for (i).